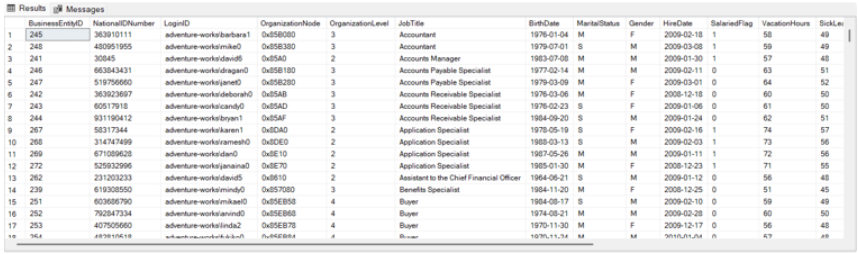
Assessment 1 - SQL

1. Write a query in SQL to retrieve all rows and columns from the employee table in the Adventureworks database. Sort the result set in ascending order on jobtitle.

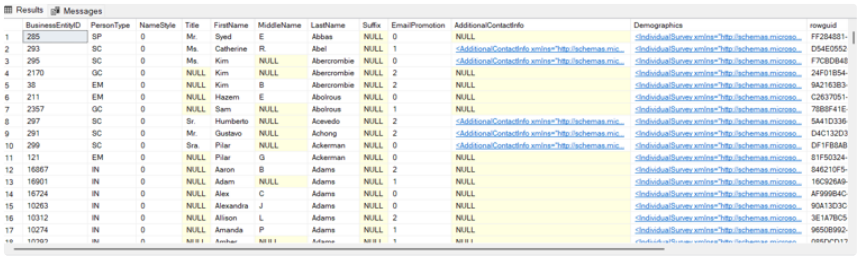
```sql
SELECT * FROM HumanResources.Employee
ORDER BY JobTitle ASC;
```



BusinessEntityID	NationalIDNumber	LoginID	OrganizationNode	OrganizationLevel	JobTitle	BirthDate	MaritalStatus	Gender	HireDate	SalariedFlag	VacationHours	SickLeaveHours
245	363910111	adventure-works\barbara1	0x85B080	3	Accountant	1976-01-04	M	F	2009-02-18	1	58	49
248	480951955	adventure-works\mike0	0x85B380	3	Accountant	1979-07-01	S	M	2009-03-08	1	59	49
241	30845	adventure-works\david5	0x85A0	2	Accounts Manager	1963-07-08	M	M	2009-01-30	1	57	48
246	663843431	adventure-works\dragan0	0x85B180	3	Accounts Payable Specialist	1977-02-14	M	M	2009-02-11	0	63	51
247	519756660	adventure-works\janet0	0x85B280	3	Accounts Payable Specialist	1979-03-09	M	F	2009-03-01	0	64	52
242	363923697	adventure-works\deborah0	0x85AB	3	Accounts Receivable Specialist	1976-03-06	M	F	2008-12-18	0	60	50
243	60517918	adventure-works\candy0	0x85AD	3	Accounts Receivable Specialist	1976-02-23	S	F	2009-01-06	0	61	50
244	931190412	adventure-works\bryan1	0x85AF	3	Accounts Receivable Specialist	1984-09-20	S	M	2009-01-24	0	62	51
267	58317344	adventure-works\karen1	0x8DA0	2	Application Specialist	1978-05-19	S	F	2009-02-16	1	74	57
268	314747499	adventure-works\ramesh0	0x8DE0	2	Application Specialist	1968-03-13	S	M	2009-02-03	1	73	56
269	671089628	adventure-works\dan0	0x8E10	2	Application Specialist	1987-05-26	M	M	2009-01-11	1	72	56
272	525932996	adventure-works\janaina0	0x8E70	2	Application Specialist	1985-01-30	M	F	2008-12-23	1	71	55
262	231203233	adventure-works\david5	0x8610	2	Assistant to the Chief Financial Officer	1964-06-21	S	M	2009-01-12	0	56	48
239	619308550	adventure-works\mindy0	0x857080	3	Benefits Specialist	1984-11-20	M	F	2008-12-25	0	51	45
251	603686790	adventure-works\mikael0	0x85EB58	4	Buyer	1984-08-17	S	M	2009-02-10	0	59	49
252	792847334	adventure-works\arvind0	0x85EB68	4	Buyer	1974-08-21	M	M	2009-02-28	0	60	50
253	407505660	adventure-works\linda2	0x85EB78	4	Buyer	1970-11-30	M	F	2009-12-17	0	56	48

2. Write a query in SQL to retrieve all rows and columns from the employee table using table aliasing in the Adventureworks database. Sort the output in ascending order on lastname.

```sql
SELECT * FROM Person.Person P
ORDER BY P.LAStName ASC;
```



BusinessEntityID	PersonType	NameStyle	Title	FirstName	MiddleName	LastName	Suffix	EmailPromotion	AdditionalContactInfo	Demographics	rowguid
285	SP	0	Mr.	Syed	E	Abbas	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso...	FF2B4B81-
293	SC	0	Ms.	Catherine	R	Abel	NULL	1	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	D54E0552
295	SC	0	Ms.	Kim	NULL	Abercrombie	NULL	0	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	F7CBDB48
2170	GC	0	NULL	Kim	NULL	Abercrombie	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso...	24F01B54-
38	EM	0	NULL	Kim	B	Abercrombie	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso...	9A2163B3-
211	EM	0	NULL	Hazem	E	Abolrous	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso...	C2637051-
2357	GC	0	NULL	Sam	NULL	Abolrous	NULL	1	NULL	<IndividualSurvey xmlns="http://schemas.microso...	76B8F41E-
297	SC	0	Sr.	Humberto	NULL	Acevedo	NULL	2	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	5A41D336-
291	SC	0	Mr.	Gustavo	NULL	Achong	NULL	2	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	D4C132D3
299	SC	0	Sra.	Pilar	NULL	Ackerman	NULL	0	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	DF1FB8AB
121	EM	0	NULL	Pilar	G	Ackerman	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso...	81F50324-
16867	IN	0	NULL	Aaron	B	Adams	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso...	846210F5-
16901	IN	0	NULL	Adam	NULL	Adams	NULL	1	NULL	<IndividualSurvey xmlns="http://schemas.microso...	16C926A9-
16724	IN	0	NULL	Alex	C	Adams	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso...	AF999B4C-
10263	IN	0	NULL	Alexandra	J	Adams	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso...	90A13D3C
10312	IN	0	NULL	Allison	L	Adams	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso...	3E1A7BC5
10274	IN	0	NULL	Amanda	P	Adams	NULL	1	NULL	<IndividualSurvey xmlns="http://schemas.microso...	9650B992-

3. Write a query in SQL to return all rows and a subset of the columns (FirstName, LastName, businessentityid) from the person table in the AdventureWorks database. The third column heading is renamed to Employee_id. Arranged the output in ascending order by lastname.

```sql
SELECT FirstName, LastName, BusinessEntityID AS Employee_ID
FROM Person.Person
ORDER BY LastName ASC;
```

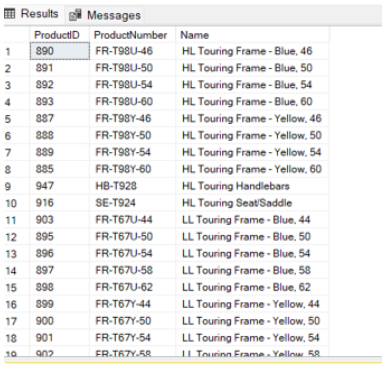


	FirstName	LastName	Employee_ID
1	Syed	Abbas	285
2	Catherine	Abel	293
3	Kim	Abercrombie	295
4	Kim	Abercrombie	2170
5	Kim	Abercrombie	38
6	Hazem	Abolrous	211
7	Sam	Abolrous	2357
8	Humberto	Acevedo	297
9	Gustavo	Achong	291
10	Pilar	Ackerman	299
11	Pilar	Ackerman	121
12	Aaron	Adams	16867
13	Adam	Adams	16901
14	Alex	Adams	16724
15	Alexandra	Adams	10263
16	Allison	Adams	10312
17	Amanda	Adams	10274
18	Amber	Adams	10292

---

4. Write a query in SQL to return only the rows for product that have a sellstartdate that is not NULL and a productline of 'T'. Return productid, productnumber, and name. Arranged the output in ascending order on name.

```sql
SELECT P.ProductID, P.ProductNumber, P.Name
FROM Production.Product P
WHERE SellStartDate IS NOT NULL AND P.ProductLine = 'T'
ORDER BY P.Name ASC;
```

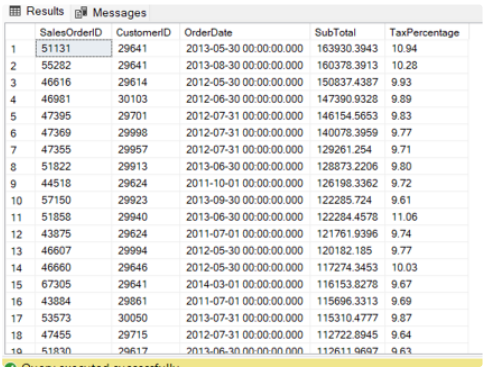


	ProductID	ProductNumber	Name
1	890	FR-T98U-46	HL Touring Frame - Blue, 46
2	891	FR-T98U-50	HL Touring Frame - Blue, 50
3	892	FR-T98U-54	HL Touring Frame - Blue, 54
4	893	FR-T98U-60	HL Touring Frame - Blue, 60
5	887	FR-T98Y-46	HL Touring Frame - Yellow, 46
6	888	FR-T98Y-50	HL Touring Frame - Yellow, 50
7	889	FR-T98Y-54	HL Touring Frame - Yellow, 54
8	885	FR-T98Y-60	HL Touring Frame - Yellow, 60
9	947	HB-T928	HL Touring Handlebars
10	916	SE-T924	HL Touring Seat/Saddle
11	903	FR-T67U-44	LL Touring Frame - Blue, 44
12	895	FR-T67U-50	LL Touring Frame - Blue, 50
13	896	FR-T67U-54	LL Touring Frame - Blue, 54
14	897	FR-T67U-58	LL Touring Frame - Blue, 58
15	898	FR-T67U-62	LL Touring Frame - Blue, 62
16	899	FR-T67Y-44	LL Touring Frame - Yellow, 44
17	900	FR-T67Y-50	LL Touring Frame - Yellow, 50
18	901	FR-T67Y-54	LL Touring Frame - Yellow, 54
19	902	FR-T67Y-58	LL Touring Frame - Yellow, 58

---

5. Write a query in SQL to return all rows from the salesorderheader table in Adventureworks database and calculate the percentage of tax on the subtotal have decided. Return salesorderid, customerid, orderdate, subtotal, percentage of tax column. Arranged the result set in ascending order on subtotal.

```sql
SELECT SalesOrderID, CustomerID, OrderDate, SubTotal,
  (TaxAmt / SubTotal) * 100 AS TaxPercentage
FROM Sales.SalesOrderHeader
ORDER BY SubTotal DESC;
```



	SalesOrderID	CustomerID	OrderDate	SubTotal	TaxPercentage
1	51131	29641	2013-05-30 00:00:00.000	163930.3943	10.94
2	55282	29641	2013-08-30 00:00:00.000	160378.3913	10.28
3	46616	29614	2012-05-30 00:00:00.000	150837.4387	9.93
4	46981	30103	2012-06-30 00:00:00.000	147390.9328	9.89
5	47395	29701	2012-07-31 00:00:00.000	146154.5653	9.83
6	47369	29998	2012-07-31 00:00:00.000	140078.3959	9.77
7	47355	29957	2012-07-31 00:00:00.000	129261.254	9.71
8	51822	29913	2013-06-30 00:00:00.000	128873.2206	9.80
9	44518	29624	2011-10-01 00:00:00.000	126198.3362	9.72
10	57150	29923	2013-09-30 00:00:00.000	122285.724	9.61
11	51858	29940	2013-06-30 00:00:00.000	122284.4578	11.06
12	43875	29624	2011-07-01 00:00:00.000	121761.9396	9.74
13	46607	29994	2012-05-30 00:00:00.000	120182.185	9.77
14	46660	29646	2012-05-30 00:00:00.000	117274.3453	10.03
15	67305	29641	2014-03-01 00:00:00.000	116153.8278	9.67
16	43884	29861	2011-07-01 00:00:00.000	115696.3313	9.69
17	53573	30050	2013-07-31 00:00:00.000	115310.4777	9.87
18	47455	29715	2012-07-31 00:00:00.000	112722.8945	9.64
19	51830	29617	2013-06-30 00:00:00.000	112611.9697	9.63

---

6. Write a query in SQL to create a list of unique jobtitles in the employee table in Adventureworks database. Return jobtitle column and arranged the resultset in ascending order.

```sql
SELECT DISTINCT(JobTitle) FROM HumanResources.Employee
ORDER BY JobTitle ASC;
```



Results Messages

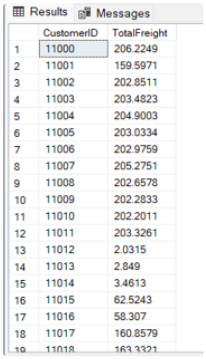
	JobTitle
1	Accountant
2	Accounts Manager
3	Accounts Payable Specialist
4	Accounts Receivable Specialist
5	Application Specialist
6	Assistant to the Chief Financial Officer
7	Benefits Specialist
8	Buyer
9	Chief Executive Officer
10	Chief Financial Officer
11	Control Specialist
12	Database Administrator
13	Design Engineer
14	Document Control Assistant
15	Document Control Manager
16	Engineering Manager
17	European Sales Manager
18	Facilities Administrative Assistant
19	Facilities Manager

Query executed successfully.

---

7. Write a query in SQL to calculate the total freight paid by each customer. Return customerid and total freight. Sort the output in ascending order on customerid.

```sql
SELECT CustomerID, SUM(Freight) AS TotalFreight
FROM Sales.SalesOrderHeader
GROUP BY CustomerID
ORDER BY CustomerID ASC;
```



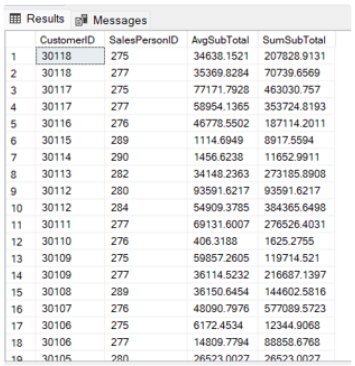
Results Messages

	CustomerID	TotalFreight
1	11000	206.2249
2	11001	159.5971
3	11002	202.8511
4	11003	203.4823
5	11004	204.9003
6	11005	203.0334
7	11006	202.9759
8	11007	205.2751
9	11008	202.6578
10	11009	202.2833
11	11010	202.2011
12	11011	203.3261
13	11012	2.0315
14	11013	2.849
15	11014	3.4613
16	11015	62.5243
17	11016	58.307
18	11017	160.8579
19	11018	163.3321

---

8. Write a query in SQL to find the average and the sum of the subtotal for every customer. Return customerid, average and sum of the subtotal. Grouped the result on customerid and salespersonid. Sort the result on customerid column in descending order.

```sql
SELECT CustomerID, SalesPersonID,
  AVG(SubTotal) AS AvgSubTotal,
  SUM(SubTotal) AS SumSubTotal
FROM Sales.SalesOrderHeader
GROUP BY CustomerID, SalesPersonID
ORDER BY CustomerID DESC;
```

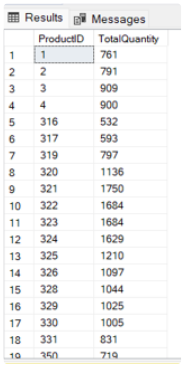


Results Messages

	CustomerID	SalesPersonID	AvgSubTotal	SumSubTotal
1	30118	275	34638.1521	207828.9131
2	30118	277	35369.8284	70739.6569
3	30117	275	77171.7928	463030.757
4	30117	277	58954.1365	353724.8193
5	30116	276	46778.5502	187114.2011
6	30115	289	1114.6949	8917.5594
7	30114	290	1456.6238	11652.9911
8	30113	282	34148.2363	273185.8908
9	30112	280	93591.6217	93591.6217
10	30112	284	54909.3785	384365.6498
11	30111	277	69131.6007	276526.4031
12	30110	276	406.3188	1625.2755
13	30109	275	59857.2605	119714.521
14	30109	277	36114.5232	216687.1397
15	30108	289	36150.6454	144602.5816
16	30107	276	48090.7976	577089.5723
17	30106	275	6172.4534	12344.9068
18	30106	277	14809.7794	88858.6768
19	30105	280	26523.0027	26523.0027

9. Write a query in SQL to retrieve total quantity of each productid which are in shelf of 'A' or 'C' or 'H'. Filter the results for sum quantity is more than 500. Return productid and sum of the quantity. Sort the results according to the productid in ascending order.

```sql
SELECT ProductID,SUM(Quantity) AS TotalQuantity
FROM Production.ProductInventory
WHERE Shelf IN ('A','C','H')
GROUP BY ProductID
HAVING SUM(Quantity) > 500
ORDER BY ProductID ASC;
```



Results Messages

	ProductID	TotalQuantity
1	1	761
2	2	791
3	3	909
4	4	900
5	316	532
6	317	593
7	319	797
8	320	1136
9	321	1750
10	322	1684
11	323	1684
12	324	1629
13	325	1210
14	326	1097
15	328	1044
16	329	1025
17	330	1005
18	331	831
19	350	719

10. Write a query in SQL to find the total quentity for a group of locationid multiplied by 10.

```sql
SELECT (Sum(Quantity)) AS TotalQuantity
FROM Production.ProductInventory
GROUP BY LocationID;
```



Results Message

	TotalQuantity
1	186
2	83173
3	17319
4	72899
5	95477
6	958
7	13584
8	110
9	20295
10	332
11	5549
12	20419
13	508
14	5165

11. Write a query in SQL to find the persons whose last name starts with letter 'L'. Return BusinessEntityID, FirstName, LastName, and PhoneNumber. Sort the result on lastname and firstname.

```sql
SELECT P.BusinessEntityID, P.FirstName, P.LastName, N.PhoneNumber
FROM Person.Person P LEFT JOIN Person.PersonPhone N ON
P.BusinessEntityID = N.BusinessEntityID
WHERE LastName LIKE 'L%'
ORDER BY LastName, P.FirstName;
```

Results Messages

	BusinessEntityID	FirstName	LastName	PhoneNumber
1	5527	Aaron	Lal	605-555-0159
2	5268	Adam	Lal	513-555-0110
3	12539	Alejandro	Lal	1 (11) 500 555-0117
4	19786	Alicia	Lal	1 (11) 500 555-0161
5	12004	Alisha	Lal	1 (11) 500 555-0119
6	16649	Alison	Lal	1 (11) 500 555-0177
7	5005	Alvin	Lal	1 (11) 500 555-0168
8	5070	Andres	Lal	1 (11) 500 555-0127
9	10416	Arturo	Lal	638-555-0164
10	8951	Ashlee	Lal	1 (11) 500 555-0148
11	6283	Austin	Lal	541-555-0141
12	11600	Barbara	Lal	1 (11) 500 555-0176
13	6744	Benjamin	Lal	1 (11) 500 555-0148
14	17275	Bethany	Lal	1 (11) 500 555-0196
15	3694	Bonnie	Lal	1 (11) 500 555-0191
16	9390	Brad	Lal	463-555-0111
17	20292	Bradley	Lal	1 (11) 500 555-0124
18	6943	Brandon	Lal	445-555-0135
19	9340	Brendan	Lal	660-555-0111

---

12. Write a query in SQL to find the sum of subtotal column. Group the sum on distinct salespersonid and customerid. Rolls up the results into subtotal and running total. Return salespersonid, customerid and sum of subtotal column i.e. sum_subtotal.

```
SELECT DISTINCT(SalesPersonID), CustomerID, SUM(Subtotal) AS
SumSubtotal
FROM Sales.SalesOrderHeader
WHERE SalesPersonID IS NOT NULL
GROUP BY SalesPersonID, CustomerID
ORDER BY SalesPersonID ASC;
```

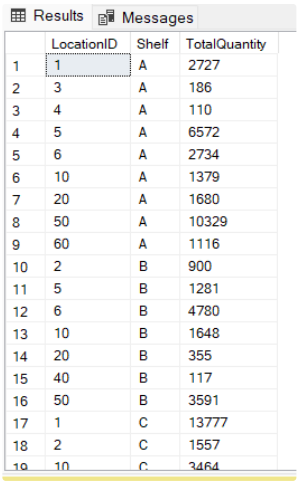
Results Messages

	SalesPersonID	CustomerID	SumSubtotal
1	274	29491	33406.7043
2	274	29493	2146.962
3	274	29514	3405.1668
4	274	29523	34349.2656
5	274	29576	53.994
6	274	29579	35331.66
7	274	29604	647.994
8	274	29605	29482.0603
9	274	29616	138046.3212
10	274	29617	198993.3507
11	274	29623	1946.022
12	274	29650	83.988
13	274	29666	15842.6141
14	274	29669	3962.2441
15	274	29671	11802.564
16	274	29675	4254.45
17	274	29680	2709.6518
18	274	29691	209.256
19	274	29707	79994.1743

---

13. Write a query in SQL to find the sum of the quantity of all combination of group of distinct locationid and shelf column. Return locationid, shelf and sum of quantity as TotalQuantity.

```
SELECT LocationID, Shelf, SUM(Quantity) AS TotalQuantity
FROM Production.ProductInventory
GROUP BY LocationID, Shelf;
```



Results Messages

	LocationID	Shelf	TotalQuantity
1	1	A	2727
2	3	A	186
3	4	A	110
4	5	A	6572
5	6	A	2734
6	10	A	1379
7	20	A	1680
8	50	A	10329
9	60	A	1116
10	2	B	900
11	5	B	1281
12	6	B	4780
13	10	B	1648
14	20	B	355
15	40	B	117
16	50	B	3591
17	1	C	13777
18	2	C	1557
19	10	C	3464

---

14. Write a query in SQL to find the sum of the quantity with subtotal for each locationid. Group the results for all combination of distinct locationid and shelf column. Rolls up the results into subtotal and running total. Return locationid, shelf and sum of quantity as TotalQuantity.

```
SELECT LocationID,Shelf, SUM(Quantity) AS TotalQuantity
FROM Production.ProductInventory
GROUP BY ROLLUP (LocationID, Shelf)
ORDER BY LocationID, Shelf;
```



Results Messages

	LocationID	Shelf	TotalQuantity
1	NULL	NULL	335974
2	1	NULL	72899
3	1	A	2727
4	1	C	13777
5	1	D	6551
6	1	E	8032
7	1	F	7614
8	1	G	3954
9	1	H	10905
10	1	J	5051
11	1	K	6751
12	1	L	7537
13	2	NULL	5549
14	2	B	900
15	2	C	1557
16	2	D	3092
17	3	NULL	186
18	3	A	186
19	4	NULL	110

---

15. Write a query in SQL to find the total quantity for each locationid and calculate the grand total for all locations. Return locationid and total quantity. Group the results on locationid.

```
SELECT LocationID, Sum(Quantity) AS TotalQuantity
FROM Production.ProductInventory
GROUP BY ROLLUP(LocationID);
```

Results | Messages

	LocationID	TotalQuantity
1	1	72899
2	2	5549
3	3	186
4	4	110
5	5	20295
6	6	83173
7	7	17319
8	10	13584
9	20	5165
10	30	958
11	40	508
12	45	332
13	50	95477
14	60	20419
15	NULL	335974

---

16. Write a query in SQL to retrieve the number of employees for each City. Return city and number of employees. Sort the result in ascending order on city.

```
SELECT City, COUNT(AddressID) AS NoOfEmployees
FROM Person.Address
GROUP BY City
ORDER BY City ASC;
```

Results | Messages

	City	NoOfEmployees
1	Abingdon	1
2	Albany	4
3	Alexandria	2
4	Alhambra	1
5	Alpine	1
6	Altadena	2
7	Altamonte Springs	1
8	Anacortes	3
9	Arlington	1
10	Ascheim	1
11	Atlanta	2
12	Auburn	1
13	Augsburg	2
14	Augusta	1
15	Aujan Mournede	1
16	Aurora	1
17	Austell	1
18	Austin	2
19	Bad Soden	1

---

17. Write a query in SQL to retrieve the total sales for each year. Return the year part of order date and total due amount. Sort the result in ascending order on year part of order date.

```
SELECT YEAR(OrderDate) AS Year, SUM(TotalDue) AS OrderAmount
FROM Sales.SalesOrderHeader
GROUP BY YEAR(OrderDate)
ORDER BY YEAR(OrderDate) ASC;
```

Results | Messages

	Year	OrderAmount
1	2011	14155699.525
2	2012	37675700.312
3	2013	48965887.9632
4	2014	22419498.3157

---

18. Write a query in SQL to retrieve the total sales for each year. Filter the result set for those orders where order year is on or before 2016. Return the year part of orderdate and total due amount. Sort the result in ascending order on year part of order date.

```
SELECT YEAR(OrderDate) AS Year,SUM(TotalDue) AS OrderAmount
```

```
FROM Sales.SalesOrderHeader
WHERE YEAR(OrderDate) <= 2016
GROUP BY YEAR(OrderDate)
ORDER BY YEAR(OrderDate) ASC;
```

Results Messages

	Year	OrderAmount
1	2011	14155699.525
2	2012	37675700.312
3	2013	48965887.9632
4	2014	22419498.3157

---

19. Write a query in SQL to find the contacts who are designated as a manager in various departments. Returns ContactTypeID, name. Sort the result set in descending order.

```
SELECT ContactTypeID, Name
FROM Person.ContactType
WHERE Name LIKE '%Manager'
ORDER BY ContactTypeID DESC;
```

Results Messages

	ContactTypeID	Name
1	19	Sales Manager
2	15	Purchasing Manager
3	13	Product Manager
4	8	Marketing Manager
5	6	International Marketing Manager
6	1	Accounting Manager

---

20. From the following tables write a query in SQL to make a list of contacts who are designated as 'Purchasing Manager'. Return BusinessEntityID, LastName, and FirstName columns. Sort the result set in ascending order of LastName, and FirstName.

```
SELECT   P.BusinessEntityID, P.FirstName, P.LastName
FROM Person.Person AS P
  JOIN Person.BusinessEntityContact AS B ON P.BusinessEntityID = B.BusinessEntityID
  JOIN Person.ContactType AS C ON B.ContactTypeID = C.ContactTypeID
WHERE C.Name = 'Purchasing Manager'
ORDER BY P.LastName ASC, P.FirstName ASC;
```

---

21. Write a query in SQL to retrieve the salesperson for each PostalCode who belongs to a territory and SalesYTD is not zero. Return row numbers of each group of PostalCode, last name, salesytd, postalcode column. Sort the salesytd of each postalcode group in descending order. Shorts the postalcode in ascending order.

```
/*SELECT TOP(5) *
FROM Sales.SalesPerson
SELECT TOP(5) *
FROM Person.Person
SELECT TOP(5) *
FROM Person.Address
SELECT TOP(5) *
FROM Person.BusinessEntityAddress*/

SELECT ROW_NUMBER() OVER(PARTITION BY A.PostalCode ORDER BY SP.SalesYTD DESC) AS RowNum,
  P.LastName, SP.SalesYTD, A.PostalCode
FROM Sales.SalesPerson AS SP
  JOIN Person.Person AS P ON SP.BusinessEntityID = P.BusinessEntityID
  JOIN Person.BusinessEntityAddress AS BEA ON P.BusinessEntityID = BEA.BusinessEntityID
  JOIN Person.Address AS A ON BEA.AddressID = A.AddressID
```

```
16 ORDER BY SP.SalesYTD DESC, A.PostalCode ASC;
```

Results Messages

	RowNum	LastName	SalesYTD	PostalCode
1	1	Mitchell	4251368.5497	84407
2	1	Pak	4116871.2277	BA5 3HX
3	1	Blythe	3763178.1787	48226
4	1	Carson	3189418.3662	55802
5	1	Varkey Chudukatil	3121616.3202	33000
6	1	Saraiva	2604540.7172	K4B 1T7
7	1	Ito	2458535.6169	94109
8	1	Reiter	2315185.611	38103
9	1	Valdez	1827066.7118	14111
10	1	Mensa-Annan	1576562.1966	02139
11	1	Campbell	1573012.9383	98004
12	1	Vargas	1453719.4653	T2P 2G8
13	1	Tsoflias	1421810.9242	3000
14	1	Ansman-Wolfe	1352577.1325	97205
15	1	Jiang	559697.5639	98052
16	1	Alberts	519905.932	98055
17	1	Abbas	172524.4512	98011

---

22. Write a query in SQL to count the number of contacts for combination of each type and name. Filter the output for those who have 100 or more contacts. Return ContactTypeID and ContactTypeName and BusinessEntityContact. Sort the result set in descending order on number of contacts.

```
/*Select * from  Person.ContactType
Select * from Person.BusinessEntityContact*/

SELECT C.ContactTypeID, C.Name, COUNT(BC.ContactTypeID) AS NoContacts
FROM Person.ContactType C JOIN Person.BusinessEntityContact BC
ON C.ContactTypeID = BC.ContactTypeID
GROUP BY C.ContactTypeID, C.Name
HAVING COUNT(BC.ContactTypeID)>=100
ORDER BY COUNT(BC.ContactTypeID) DESC
```

Results Messages

	ContactTypeID	Name	NoContacts
1	11	Owner	266
2	15	Purchasing Manager	245
3	14	Purchasing Agent	242

---

23. Write a query in SQL to retrieve the RateChangeDate, full name (first name, middle name and last name) and weekly salary (40 hours in a week) of employees. In the output the RateChangeDate should appears in date format. Sort the output in ascending order on NameInFull.

```
SELECT FORMAT(E.RateChangeDate, 'yyyy-MM-dd') AS FromDate,
  CONCAT(  P.FirstName, ', ', P.MiddleName, ' ', LEFT(P.LastName,1)) AS
NameInFull,
  E.Rate * 40 AS SalaryInAWeek
FROM HumanResources.EmployeePayHistory AS E
  JOIN Person.Person AS P ON E.BusinessEntityID = P.BusinessEntityID
ORDER BY NameInFull ASC;
```

Results Messages

	FromDate	NameInFull	SalaryInAWeek
1	2008-12-12	A. Scott, W	942.308
2	2009-02-13	Alan, J B	640.00
3	2008-12-06	Alejandro, E M	600.00
4	2009-02-08	Alex, M N	400.00
5	2008-12-07	Alice, O C	440.00
6	2012-04-16	Amy, E A	1924.04
7	2009-02-02	Andreas, T B	423.076
8	2009-03-06	Andrew, M C	400.00
9	2009-02-22	Andrew, R H	1000.00
10	2009-01-31	Andy, M R	380.00
11	2009-01-20	Angela, W B	440.00
12	2009-02-23	Anibal, T S	560.00
13	2010-12-06	Annette, L H	510.00
14	2008-12-17	Annik, O S	498.00
15	2009-02-28	Arvind, B R	730.768
16	2008-12-04	Ashvini, R S	1298.076
17	2009-02-18	Barbara, C M	1057.692
18	2009-01-22	Barbara, S D	560.00
19	2009-02-15	Baris, F C	600.00

---

24. Write a query in SQL to calculate and display the latest weekly salary of each employee. Return RateChangeDate, full name (first name, middle name and last name) and weekly salary (40 hours in a week) of employees Sort the output in ascending order on NameInFull.

```
SELECT * FROM(
  SELECT FORMAT(E.RateChangeDate, 'yyyy-MM-dd') AS FromDate,
    CONCAT(  P.FirstName, ', ', P.MiddleName, ' ', LEFT(P.LastName,1))
AS NameInFull,
    E.Rate * 40 AS SalaryInAWeek,
    ROW_NUMBER() OVER( PARTITION BY E.BusinessEntityID ORDER BY
E.RateChangeDate DESC) AS RN
  FROM HumanResources.EmployeePayHistory AS E
    JOIN Person.Person AS P ON E.BusinessEntityID = P.BusinessEntityID)
AS PAY
WHERE RN = 1
ORDER BY NameInFull ASC;
```

Results Messages

	FromDate	NameInFull	SalaryInAWeek
1	2008-12-12	A. Scott, W	942.308
2	2009-02-13	Alan, J B	640.00
3	2008-12-06	Alejandro, E M	600.00
4	2009-02-08	Alex, M N	400.00
5	2008-12-07	Alice, O C	440.00
6	2012-04-16	Amy, E A	1924.04
7	2009-02-02	Andreas, T B	423.076
8	2009-03-06	Andrew, M C	400.00
9	2009-02-22	Andrew, R H	1000.00
10	2009-01-31	Andy, M R	380.00
11	2009-01-20	Angela, W B	440.00
12	2009-02-23	Anibal, T S	560.00
13	2010-12-06	Annette, L H	510.00
14	2008-12-17	Annik, O S	498.00
15	2009-02-28	Arvind, B R	730.768
16	2008-12-04	Ashvini, R S	1298.076
17	2009-02-18	Barbara, C M	1057.692
18	2009-01-22	Barbara, S D	560.00
19	2009-02-15	Baris, F C	600.00

---

25. Write a query in SQL to find the sum, average, count, minimum, and maximum order quentity for those orders whose id are 43659 and 43664. Return SalesOrderID, ProductID, OrderQty, sum, average, count, max, and min order quantity.

```
SELECT S.SalesOrderID, S.ProductID, S.OrderQty,
  SUM(S.OrderQty) OVER() AS TotalQuantity,
  AVG(S.OrderQty) OVER() AS AvgQuantity,
  COUNT(S.OrderQty) OVER()  AS NoOfOrders,
  MIN(S.OrderQty) OVER() AS MinQuanity,
  MAX(S.OrderQty) OVER() AS MaxQuantity
FROM Sales.SalesOrderDetail S
```

```
8 WHERE S.SalesOrderID IN (43659, 43664)
```

Results Messages

	SalesOrderID	ProductID	OrderQty	TotalQuantity	AvgQuantity	NoOfOrders	MinQuanity	MaxQuantity
1	43659	776	1	40	2	20	1	6
2	43659	777	3	40	2	20	1	6
3	43659	778	1	40	2	20	1	6
4	43659	771	1	40	2	20	1	6
5	43659	772	1	40	2	20	1	6
6	43659	773	2	40	2	20	1	6
7	43659	774	1	40	2	20	1	6
8	43659	714	3	40	2	20	1	6
9	43659	716	1	40	2	20	1	6
10	43659	709	6	40	2	20	1	6
11	43659	712	2	40	2	20	1	6
12	43659	711	4	40	2	20	1	6
13	43664	772	1	40	2	20	1	6
14	43664	775	4	40	2	20	1	6
15	43664	714	1	40	2	20	1	6
16	43664	716	1	40	2	20	1	6
17	43664	777	2	40	2	20	1	6
18	43664	771	3	40	2	20	1	6
19	43664	773	1	40	2	20	1	6